

LP5 Practical Viva Preparation Guide

SPPU Computer Engineering – Final Year Sem 8

Subjects: High Performance Computing (HPC) and Deep Learning (DL)

Part A: High Performance Computing (HPC)

1. What is High Performance Computing?

High Performance Computing (HPC) is the use of parallel processing and powerful computers to solve large and complex problems faster.

One-line answer:

HPC uses multiple processors or GPUs to perform computations quickly.

2. Why do we need HPC?

- To solve very large problems.
 - To reduce execution time.
 - To process huge amounts of data.
 - Used in weather forecasting, AI, simulations, and scientific research.
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3. What is Parallel Computing?

Parallel computing means dividing a task into smaller parts and executing them simultaneously.

Example:

If 1000 additions are needed:

- Sequential: one by one.
 - Parallel: many additions at the same time.
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4. Sequential vs Parallel Computing

Sequential	Parallel
One task at a time	Many tasks at once
Slower	Faster
Single processor	Multiple processors/threads

5. What is OpenMP?

OpenMP is an API for parallel programming in C/C++.

One-line answer:

OpenMP allows us to parallelize loops using compiler directives.

6. What is CUDA?

CUDA is a GPU programming platform developed by NVIDIA.

One-line answer:

CUDA allows C/C++ programs to run on the GPU.

7. CPU vs GPU

CPU	GPU
Few powerful cores	Thousands of smaller cores
Good for general tasks	Good for parallel tasks
Sequential processing	Massive parallel processing

HPC Practical-Wise Viva Questions

Experiment 1: Parallel BFS

What is BFS?

Breadth First Search traverses a graph level by level.

Which data structure is used?

Queue.

Time Complexity

$O(V + E)$

Space Complexity

$O(V + E)$

Can BFS find shortest path?

Yes, in unweighted graphs.

Why use visited array?

To avoid visiting the same node multiple times.

One-line answer:

BFS uses a queue to visit all nodes level by level.

Experiment 2: Parallel DFS

What is DFS?

Depth First Search explores one path completely before backtracking.

Which data structure is used?

Stack.

Time Complexity

$O(V + E)$

Space Complexity

$O(V + E)$

One-line answer:

DFS uses a stack to explore nodes deeply before backtracking.

Experiment 3: Parallel Bubble Sort

What is Bubble Sort?

Repeatedly compares adjacent elements and swaps them if needed.

Time Complexity

$O(n^2)$

Space Complexity

$O(1)$

Is Bubble Sort stable?

Yes.

What is Odd-Even Transposition Sort?

A parallel-friendly version of Bubble Sort.

One-line answer:

Bubble Sort repeatedly swaps adjacent elements until the array is sorted.

Experiment 4: Parallel Merge Sort

What is Merge Sort?

A divide-and-conquer sorting algorithm.

Steps

1. Divide
2. Sort recursively
3. Merge

Time Complexity

$O(n \log n)$

Space Complexity

$O(n)$

Is Merge Sort stable?

Yes.

One-line answer:

Merge Sort divides the array, sorts each half, and merges them.

Experiment 5: Parallel Reduction

What is Reduction?

Combining multiple values into one.

Examples:

- Min
- Max
- Sum
- Average

Time Complexity

$O(n)$

Why use long long for sum?

To avoid overflow.

One-line answer:

Reduction combines many elements into one result.

Experiment 6: CUDA Vector Addition

What is a Vector?

A one-dimensional array.

What does each CUDA thread do?

Adds one element.

Time Complexity

$O(n)$

Space Complexity

$O(n)$

One-line answer:

Each GPU thread adds one pair of vector elements.

Experiment 7: CUDA Matrix Multiplication

What does each thread compute?

One element of the result matrix.

Formula

$C[i][j] = \sum(A[i][k] * B[k][j])$ for all k .

Time Complexity

$O(n^3)$

Space Complexity

$O(n^2)$

One-line answer:

Each GPU thread computes one element of the result matrix.

Important OpenMP Questions

What is #pragma omp parallel for?

It distributes loop iterations among multiple threads.

What is omp_get_wtime()?

Returns the current wall-clock time.

What is speedup?

Speedup = Sequential Time / Parallel Time

What is a thread?

A lightweight execution unit.

What is a race condition?

When multiple threads access shared data simultaneously causing incorrect results.

What is a critical section?

A block where only one thread executes at a time.

What is reduction clause?

Safely combines thread-local values into one final result.

Important CUDA Questions

What is a Kernel?

A function that runs on the GPU.

What does global mean?

The function runs on GPU and is called from CPU.

What is a thread?

Smallest execution unit on GPU.

What is a block?

Group of threads.

What is a grid?

Group of blocks.

What is Host?

CPU memory.

What is Device?

GPU memory.

What does cudaMalloc() do?

Allocates memory on the GPU.

What does cudaMemcpy() do?

Copies data between CPU and GPU.

What does cudaFree() do?

Frees GPU memory.

Part B: Deep Learning Viva Questions

1. What is Artificial Intelligence (AI)?

AI is the field of creating systems that mimic human intelligence.

2. What is Machine Learning (ML)?

ML enables systems to learn patterns from data.

3. What is Deep Learning (DL)?

Deep Learning uses neural networks with multiple hidden layers.

Relationship

$AI \supset ML \supset DL$

4. What is a Neural Network?

A network of interconnected neurons that learns patterns from data.

5. What is a Neuron?

A computational unit that: 1. Takes inputs. 2. Multiplies by weights. 3. Adds bias. 4. Applies activation. 5. Produces output.

6. What are Weights?

Values that determine the importance of each input.

7. What is Bias?

A constant added to shift the output.

8. What is an Activation Function?

Introduces non-linearity.

Examples:

- ReLU
 - Sigmoid
 - Tanh
 - Softmax
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9. What is ReLU?

$\text{ReLU}(x) = \max(0, x)$

Advantages:

- Simple
 - Fast
 - Avoids vanishing gradient
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10. What is Sigmoid?

Outputs values between 0 and 1.

Used for binary classification.

11. What is Softmax?

Converts outputs into probabilities.

Used for multi-class classification.

12. What is Forward Propagation?

Computing outputs from inputs.

13. What is Loss Function?

Measures prediction error.

Examples:

- Mean Squared Error
 - Cross Entropy
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14. What is Backpropagation?

Computes gradients and updates weights.

15. What is Gradient Descent?

Optimization method used to minimize loss.

16. What is Learning Rate?

Controls how much weights are updated.

Too small → slow. Too large → unstable.

17. What is an Epoch?

One complete pass over the training dataset.

18. What is Batch Size?

Number of samples processed before updating weights.

19. What is Iteration?

One update step.

Iterations per epoch = Number of samples / Batch size.

20. What is Overfitting?

Model performs well on training data but poorly on new data.

Solutions

- Dropout
 - Regularization
 - More data
 - Early stopping
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21. What is Underfitting?

Model fails to learn patterns.

22. What is Dropout?

Randomly disables neurons during training.

23. What is Regularization?

Penalizes large weights to reduce overfitting.

24. What is CNN?

Convolutional Neural Network used for images.

Applications:

- Face recognition
 - Object detection
 - Medical imaging
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25. What is Convolution?

Applying a filter to extract features.

26. What is Filter (Kernel in CNN)?

A small matrix used to detect patterns.

27. What is Pooling?

Reduces feature map size.

Types:

- Max Pooling
 - Average Pooling
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28. What is Flatten?

Converts a 2D feature map into a 1D vector.

29. What is Dense Layer?

Fully connected layer.

30. What is RNN?

Recurrent Neural Network for sequential data.

Applications:

- Text
 - Speech
 - Time series
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31. What is LSTM?

A type of RNN that remembers long-term dependencies.

32. What is GRU?

Simplified version of LSTM.

33. What is Transfer Learning?

Using a pre-trained model and fine-tuning it.

34. What is Data Augmentation?

Generating additional training data by modifying existing images.

Examples:

- Rotation
 - Flip
 - Zoom
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35. What is Confusion Matrix?

Shows counts of correct and incorrect predictions.

36. What is Accuracy?

Correct predictions / Total predictions.

37. What is Precision?

True Positives / (True Positives + False Positives)

38. What is Recall?

True Positives / (True Positives + False Negatives)

39. What is F1 Score?

Harmonic mean of Precision and Recall.

40. What is TensorFlow?

An open-source deep learning framework by `entity["company","Google","AI framework developer"]`.

41. What is Keras?

A high-level API used to build neural networks easily.

42. What is PyTorch?

A deep learning framework developed by `entity["company","Meta Platforms","AI framework developer"]`.

43. What is a Tensor?

A multidimensional array.

44. What is Feature Extraction?

Finding useful patterns from data.

45. What is Classification?

Predicting categories.

Example: Cat or Dog.

46. What is Regression?

Predicting continuous values.

Example: House price.

47. What is Label?

Correct output value.

48. What is Training Data?

Data used to train the model.

49. What is Validation Data?

Data used to tune the model.

50. What is Test Data?

Data used to evaluate final performance.

Common Deep Learning Practical Questions

Why use ReLU?

It is simple, fast, and helps avoid vanishing gradients.

Why use Softmax in the output layer?

It converts outputs into class probabilities.

Why normalize data?

It speeds up and stabilizes training.

Why split data into train and test sets?

To evaluate generalization.

What optimizer is commonly used?

Adam.

What loss is used for binary classification?

Binary Crossentropy.

What loss is used for multi-class classification?

Categorical Sparse Crossentropy.

Most Important Viva Questions (Must Know)

1. What is AI?
2. What is ML?
3. What is Deep Learning?

4. What is a neural network?
 5. What are weights and bias?
 6. What is an activation function?
 7. What is ReLU?
 8. What is backpropagation?
 9. What is gradient descent?
 10. What is learning rate?
 11. What is epoch?
 12. What is batch size?
 13. What is overfitting?
 14. What is dropout?
 15. What is CNN?
 16. What is pooling?
 17. What is accuracy?
 18. What is TensorFlow?
 19. What is Keras?
 20. What is transfer learning?
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Quick 1-Line Answers for Viva

- OpenMP: API for parallel programming in C/C++.
 - CUDA: GPU programming platform by NVIDIA.
 - BFS: Level-order graph traversal using queue.
 - DFS: Depth-first graph traversal using stack.
 - Bubble Sort: Adjacent swapping sort.
 - Merge Sort: Divide and conquer sorting.
 - Reduction: Combining many values into one.
 - Neural Network: Layers of neurons that learn patterns.
 - ReLU: $\max(0, x)$.
 - Epoch: One complete pass over training data.
 - CNN: Neural network for image processing.
 - Overfitting: Good on training data, poor on test data.
 - Dropout: Randomly disables neurons during training.
 - Accuracy: Correct predictions divided by total predictions.
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Final Practical Tips

For HPC

- Know Queue, Stack, Sorting, Reduction, CUDA basics.
- Understand time and space complexity.
- Be able to explain each function.

For Deep Learning

- Understand neural networks.
- Know activation functions.
- Know CNN basics.
- Understand overfitting and dropout.

If You Don't Know an Answer

Say:

"I am not fully sure, but my understanding is..."

Final Confidence Statements

For HPC

"This program compares sequential and parallel approaches to improve performance."

For Deep Learning

"Deep Learning uses neural networks with multiple hidden layers to automatically learn patterns from data."

Best of Luck!

You already have the practical codes and explanations. Review these questions, stay calm, and answer confidently.